

Assignment: Exploratory Data Analysis with NumPy, Pandas, and

Matplotlib Title: Student Performance Analysis Using Python

Objective

In this assignment, you will use **NumPy**, **Pandas**, and **Matplotlib** to analyze a real-world dataset. You will clean the data, perform statistical analysis, answer business questions, and visualize your findings.

Dataset

Students Performance Dataset (Kaggle)

- **Link:** <https://www.kaggle.com/datasets/spscientist/students-performance-in-exams>

Alternative (CSV directly):

- <https://raw.githubusercontent.com/mwaskom/seaborn-data/master/exercise.csv>
(simpler but less suitable)

For this assignment, use the **Students Performance Dataset**.

Learning Outcomes

By completing this assignment, students should be able to:

- Load datasets using Pandas
- Explore and summarize data
- Perform numerical operations using NumPy
- Clean and preprocess data
- Create informative visualizations using Matplotlib
- Draw conclusions from data

Part A – Loading and Exploring Data

Load the dataset using Pandas.

1. Display:
 - First 5 rows
 - Last 5 rows
2. Find:
 - Number of rows
 - Number of columns
3. Display all column names.
4. Check data types of every column.

Part B – Data Cleaning

1. Check whether the dataset contains missing values.
2. Remove duplicate rows (if any).
3. Verify that the dataset is clean.

Part C – NumPy Analysis

Convert the following columns into NumPy arrays:

- math score
- reading score
- writing score

Using NumPy, calculate:

1. Mean
2. Median
3. Maximum
4. Minimum
5. Standard Deviation

Also calculate:

- Average score of each student

Create a new Pandas column called:

Average Score

Part D – Data Analysis with Pandas

Answer the following questions.

1. Which student achieved the highest average score?
2. Which subject has the highest average marks?
3. How many students scored above 90 in Mathematics?
4. Find the average score grouped by:
 - Gender
5. Find the average score grouped by:
 - Parental Level of Education
6. Which lunch type resulted in higher average marks?
7. Compare students who completed the test preparation course versus those who did not.

Part E – Visualization

Using **Matplotlib only**, create the following plots.

1. Histogram

Distribution of Math Scores

2. Bar Chart

Average score in:

- Math
- Reading
- Writing

3. Scatter Plot

Reading Score vs Writing Score

4. Pie Chart

Distribution of Students by Gender

5. Line Plot

Average score of every student (sorted from lowest to highest).

Part F – Interpretation

Write short answers (2–4 sentences each).

1. Which subject appears easiest for students?
2. Does test preparation improve performance?
3. Which visualization provided the most useful insight?
4. Suggest one improvement that schools could make based on the data.

Submission Requirements

Submit:

- Python Notebook (.ipynb)
- PDF Report
- All graphs properly labeled with:
 - Title
 - X-axis
 - Y-axis
- Comment your code where necessary.